Attention, Perception, & Psychophysics
https://doi.org/10.3758/s13414-024-02873-y

Suppression on the basis of template for rejection is reactive: Evidence from human electrophysiology

Chao Pang[1] · Yanzhang Chen[2] · Yue Zhang[1] · Weizhi Nan[1] · Shimin Fu[1]

Accepted: 21 February 2024

Abstract
According to most theories of attention, the selection of task-relevant visual information can be enhanced by holding them in visual working memory (VWM). However, there has been a long-standing debate concerning whether similar optimization can also be achieved for task-irrelevant information, known as a "template for rejection". The present study aimed to explore this issue by examining the consequence of cue distractors before visual search tasks. For this endeavor, we manipulated the display heterogeneity by using two distractor conditions, salient and non-salient, to explore the extent to which holding the distractor color in VWM might affect attentional selection. We measured the reaction times of participants while their EEG activity was recorded. The results showed that WM-matched distractors did not improve reaction times but rather slowed them down in both tasks. Event-related potential (ERP) results showed that the display heterogeneity had no modulatory effect on the degree of distractor suppression. Even in the salient distractor condition, the WM-matched distractor received no greater suppression. Furthermore, the WM-matched distractor but not the neutral distractor elicited an N2pc before the P_D in salient distractor conditions. This suggests that the template for rejection operates reactively since suppression occurs after extra attentional processes to the distractor. Moreover, the presence of WM-matched distractors led to a reduction of P3b, indicating a competition between target processing and WM-matched distractor rejection. Our findings provide insights into the mechanisms underlying the optimization of attentional selection, and have implications for future studies aimed at understanding the role of VWM in cognition.

Keywords Visual search · Visual working memory · Template for rejection

Introduction

The ability to identify one object among others is fundamental for humans. To do so, a person may need to know the object-relative information to constrain the attentional selection while determining the relevance of information for search processes. This cognitive process was assumed by most theories of attention, controlled by the mental representation known as the attentional template (Duncan & Humphreys, 1992; Wolfe, 2012) or attentional control sets (Folk et al., 1992) that are actively maintained in visual working memory (VWM). The way in which VWM biases our attention has long been of interest in visual search. Despite a vast body of research that has sought to test how attentional selection is optimized by task-relevant representations, there is a long-running debate over whether such optimization can be achieved by representations that are known to be irrelevant.

The research of Woodman and Luck (2007) provided compelling evidence for the idea that VWM content can serve as a "template for rejection", rather than always biasing attention. Their study revealed that manual responses were faster when a VWM-matched distractor was present instead of absent, indicating that it could be utilized to optimize attentional selections (see also Arita et al., 2012; Carlisle & Nitka, 2019). In recent years, researchers have

Chao Pang, Yanzhang Chen and Yue Zhang contributed equally to this research.

Shimin Fu
fusm@gzhu.edu.cn

[1] Department of Psychology and Center for Brain and Cognitive Sciences, School of Education, Guangzhou University, Guangzhou 510006, China

[2] Philosophy and Social Science Laboratory of Reading and Development in Children and Adolescents (South China Normal University), Ministry of Education, Guangzhou, China

sought to explore the underlying mechanisms of how task-irrelevant VWM promotes certain responses (Addleman & Störmer, 2022; Beck et al., 2018; Carlisle, 2023; Carlisle & Nitka, 2019; Zhang et al., 2020). Numerous studies consistently reported attentional selection is biased towards VWM-matched objects, even when these objects were intended to be avoided (Addleman & Störmer, 2022; Beck et al., 2018; Gaspelin & Luck, 2019; Zhang & Carlisle, 2023). However, a small but significant body of research has presented conflicting results, suggesting that holding the distractor feature in VWM can optimize attentional selections under specific conditions. For example, in the situation where the VWM-matched distractor reappears in the search task on each trial (Berggren & Eimer, 2021; van Moorselaar et al., 2016) or during an effortful search task (Conci et al., 2019; Han & Kim, 2009) has yielded different results. To summarize, the concept of a "template for rejection" is currently surrounded by controversy, and it becomes imperative to delve deeper into the investigation and exploration of its underlying causes.

To filter out task-irrelevant distractors, two methods are commonly implemented: reactive suppression and active suppression. Reactive suppression involves the rapid disengagement of attentional responses towards distractors using cognitive control mechanisms, while active suppression directly diverts attention away from distractors, preventing any capture of attention. Active suppression appears to offer a more efficient solution to the problem of distractor rejection, and theoretical models suggest that the use of a template for rejection can optimize search performance by suppressing attentional responses towards matching distractors before their appearance. However, this view seems paradoxical when considered in the context of the biased-competition theory (Desimone & Duncan, 1995). According to this theory, guided attention from VWM operates in a top-down manner, directing attentional selection towards a visual input that matches the contents of VWM. In situations where a cued feature matches a distractor, the activated representations of distractor features can interact with subsequent sensory processing, biasing visual attention towards the matched distractor. To avoid such bias, reactive control mechanisms may be employed to swiftly disengage prepared attentional responses towards distractors. This reactive mechanism is consistent with the "search-and-destroy" account proposed by Moher and Egeth (2012), which entails explicit suppression following the selection of a to-be-ignored color.

Studies have shown that rejecting distractors based on matching VWM contents can result in faster responses compared to trials with no VWM-matched distractors, indicating that rapid distractor rejection can reduce the impact of matching distractors (Beck et al., 2015, 2018 but see Theeuwes, 2010, for a bottom-up account of this prediction). However, the theory also suggests that bottom-up processes may influence distractor rejection, providing another approach to this problem (Theeuwes, 2010). In an eye-tracking study, Beck et al. (2018) found that if the VWM-matched distractor captures the early fixation, there was no subsequent suppression on it. But if the VWM-matched distractor failed to capture the early fixation, later suppression occurred (Carlisle, 2023). This suggests that initial attention to VWM-matched distractors may interfere with later suppression. A recent review by Gaspelin and Luck (2019) suggested that proactive suppression cannot be built up on explicit goals, which refers to the situation that holding a distractor feature actively in VWM is insufficient to directly reject the matched item (for a more detailed review, see also Luck et al., 2021, p.8, line 23). In line with this opinion, recent studies have revealed that suppression over the VWM-matched distractor is rather reactive, with visual attention initially deployed to the to-be-ignored location, followed by subsequent suppression of it (Addleman & Störmer, 2022; Gaspelin & Luck, 2019). Overall, the operation of a template for rejection is expected to be reactive and engage quickly to interrupt biased attentional selection towards distractors.

To investigate the reactive suppression account, we analyzed the N2pc and P_D components, which are event-related potential (ERP) signatures of visuo-spatial attention deployment. The N2pc is a transient negativity enhancement evident in a 200- to 300-ms time-window at parieto-occipital sites contralateral to the side where the target is displayed (Eimer, 1996; Luck & Hillyard, 1994). In contrast, the P_D is detected at the same recording sites as the N2pc but manifests as a larger positivity contralateral to the side where the distractor to be suppressed is displayed (Hickey, Di Lollo, & McDonald, 2008). Additionally, the non-lateralized component P3b was measured, which reflects the updating of information in VWM (Cosman et al., 2016; Polich, 2007; Vogel & Luck, 2002). A recent study suggested that an enhanced P3b amplitude may also represent a bias towards categorization and decision-making (Rac-Lubashevsky & Kessler, 2019). To answer the research question of whether the WM-matched distractor elicited a larger P3b than the neutral distractor, we compared the P3b amplitudes between these conditions. If the WM-matched distractor induces similar P3b amplitude to the neutral distractor, it suggests that the WM-matched distractor does not consume limited-capacity resources such as target identification during visual search (Peters et al., 2008).

It is important to acknowledge that the search strategy used in experiments can impact how efficiently distractors are suppressed (for a review, see Gaspelin & Luck, 2018b; Luck et al., 2021). For example, when searching for a target with a specific feature value among heterogenous non-targets (known as *feature-search mode*), attentional

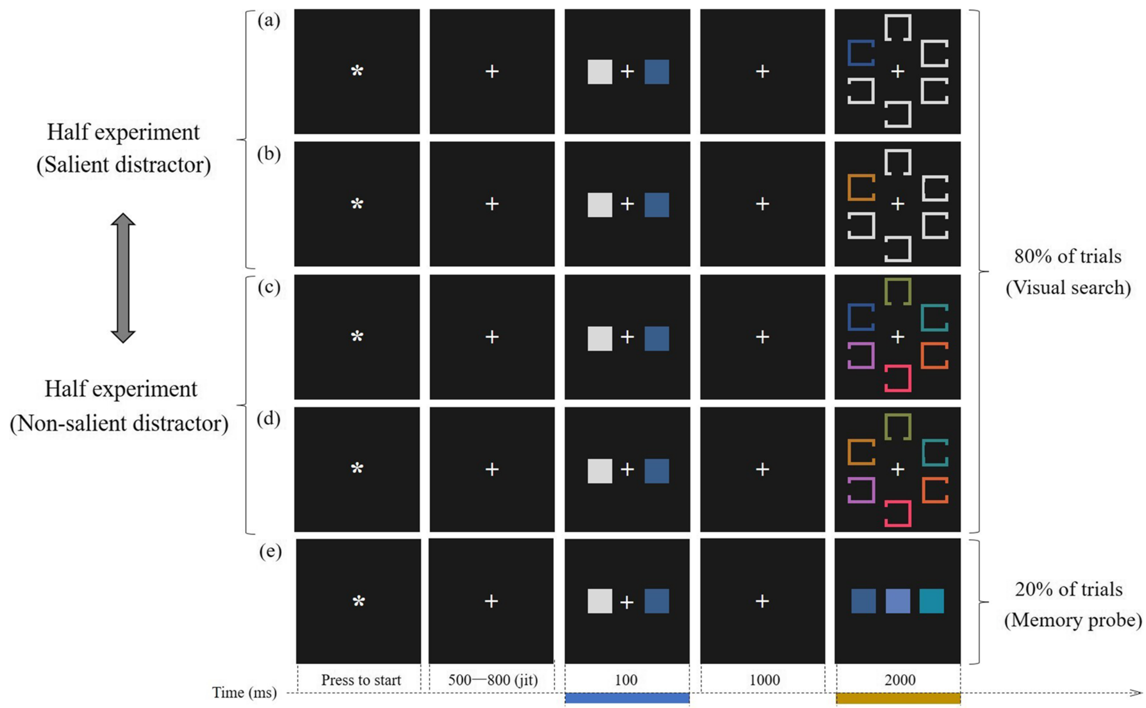

Fig. 1 Schematic illustration of task design. Participants were given one color to remember (always paired with the white square), after which they performed either the visual search task (in 80% of trials, **a** to **d**) or the memory probe test (in 20% of trials, **e**). In the salient distractor task (**a** and **b**) the search display contained one colored square that never matched any target features, whereas all items differed in colors in the non-salient distractor task (**c** and **d**). The order of task type was counterbalanced across participants. In both tasks, one of the search distractors could match the color of the memory cue (WM-matched trials, **a** and **c**). On neutral trials, there was no match between the cued color and any colors in the subsequent search task (**b** and **d**)

selection is less susceptible to capture by physically salient distractors (Bacon & Egeth, 1994; Burra & Kerzel, 2013; Gaspelin & Luck, 2018a; Leber & Egeth, 2006). The underlying assumption of this observation was interpreted as display heterogeneity reducing the physical saliency of the distractor (Gaspelin & Luck, 2018b; for a review, see Luck et al., 2021). However, we cannot simply analogize the above situation to memory-driven capture, because the memory-matched item does not inherently possess bottom-up saliency relative to other search items. In other words, "template for rejection" refers to suppressing a memory-matched but physically non-salient distractor by means of online memory representations. The issue at stake in the present context is whether memory-driven suppression is contingent on search strategy. To address this question, we investigated the reactive suppression account while participants were cued to remember the distractor color before the search task. We manipulated display heterogeneity (salient vs. non-salient distractors) to accomplish this goal, which is explained further below.

Participants in the present study completed two types of visual search tasks. The order of the tasks was counterbalanced, as shown in Fig. 1. During both tasks, participants had to actively maintain a colored cue related to one of the search distractors in VWM. In the *salient distractor task,* all search items were white except for one distractor item that always appeared in a singleton color. This provided a bottom-up inherent saliency "signal" that would lead to an "attend-to-me" bias (Sawaki & Luck, 2010). When encouraged to be in a "feature-search mode", the singleton distractor was discouraged from capturing attention (Bacon & Egeth, 1994). In half of the trials, the singleton distractor shared the same color as the memory color. We hypothesized that if the content of WM can be configured as a template for rejection, the singleton distractor related to the content of WM should receive stronger suppression than if it were unrelated.

In contrast, the *non-salient distractor task* examined whether the degree of suppression was reduced when no search items popped out from the display due to display heterogeneity. The "search-and-destroy" account predicts that participants first deploy their attention to the to-be-ignored distractor and then actively suppress whatever appears in that location. As such, the lateral singleton would elicit an N2pc first as attention was initially directed toward its location, followed by a suppression-related P_D as attention was directed away from it. The active suppression account, on the other hand, predicts only suppression-related P_D would be observed because suppression of the known irrelevant feature reacts before the search display onset. The reactive suppression account predicts spatial certainty of the WM-matched distractor is necessary before rejection. Therefore, suppression over the WM-matched distractor is expected to occur more strongly and/or faster in the salient relative to the non-salient distractor task.

Method

Participants

Twenty-five students at Guangzhou University (three males; mean age = 19 years, *SD* = 1.6) took part in the present experiment after providing written informed consent. All participants had normal or corrected-to-normal visual acuity, and all reported normal color vision and no history of neurological disorders. The experiment was vetted by the local ethics committee.

Stimuli and procedure

An example of the stimuli and a schematic illustration of the sequence of events on four trials in the experiment are shown in Fig. 1. The stimuli were displayed on the black background (CIE: 0.312/0.329, 1.0 cd/m^2) of a 17-in. CRT computer monitor with a refresh rate of 60 Hz, at a viewing distance of about 75 cm. The stimuli in the cue array (marked by the cyan bar on the timeline in Fig. 1) were two squares (2° × 2°), symmetrically located at 3.2° of visual angle on the left and right of fixation. One colored square represented the WM items, whereas the other one was a task-irrelevant white square (CIE: 0.313/0.329, 90 cd/m^2). The search arrays (marked by the orange bar on the timeline in Fig. 1) were composed of six outlined squares, arranged along a notional circle of 4.4° in diameter and positioned in correspondence to the number of locations on a clock face. The target had a top or bottom gap (0.7°), whereas distractors had a left or right gap. Colors for the WM items in the cue array and the stimuli in the search array were equiluminant (20 cd/m^2), randomly selected from seven possible values (i.e., CIE: 0.276/0.381; 0.214/0.254; 0.256/0.246; 0.355/0.231; 0.500/0.287; 0.526/0.388; 0.400/0.452; Zhang & Luck, 2008).

Each trial began with the presentation of a white fixation dot (0.6° × 0.6°) at the center of the monitor. Participants were instructed to maintain their gaze on the fixation dot, avoiding head and/or eye movements until the end of the trial. Participants started each trial by pressing the spacebar using the thumb of the left or right hand. After the spacebar press, an interval of 500–800 ms (randomly jittered using a rectangular distribution) elapsed before the onset of the cue array, which was exposed for 100 ms. The colored square in the cue array could unpredictably and with equal probability be displayed to the left or right of the central fixation cross. The color of the square in the cue array had to be memorized regardless of its spatial arrangement. Participants were told that the cue would always match the distractor color. They could strategically use this information to promote their search performance. After a 1,000-ms blank screen, the search array was exposed for 200 ms. The maximum time for responding was 1,800 ms. Participants were asked to report whether the target had a gap on its top or bottom by pressing one of the two response keys with the index or middle fingers of their right hand (i.e., "Y" or "B"), with equal emphasis on response speed and accuracy. Participants could be exposed to two different search arrays (with the order counterbalanced across participants): salient distractor task (Fig. 1a and b), in which five search items were color uniform (CIE: 0.313/0.329, 90 cd/m^2), with the addition of one distractor of gapped squares that was color singleton; non-salient distractor task (Fig. 1c and d), in which the target and distractors differed in colors.

In the WM-matched trials, one distractor was presented in the color that was cued (in salient distractor trials it was the color singleton). Locations of search items were varied pseudo-randomly to obtain ERPs time-locked to search targets and WM-matched distractors.

On one-third of WM-matched trials, the target appeared aligned with the vertical meridian (i.e., the positions at 12 and 6 o'clock), while the WM-matched distractor appeared laterally (i.e., left/right); on the other two-thirds of trials, their relative positions were reversed. To create neutral trials that mirror the WM-matched trials, the cued color did not appear in the search array. Memory probes were given on 20% of trials, rather than the search array, to prevent participants from strategically using it to refresh their WM of the cued color during the search task (strategic perceptual resampling hypothesis; Woodman & Luck, 2007), so that participants had no motivation for attending to the WM-matched distractor (see Dowd et al., 2015, for the same manipulation). Participants were asked to choose the remembered color from three options by pressing one of the three response keys with the left index, middle, or ring finger, respectively (i.e., "D", "F", or "G").

The experiment consisted of 12 blocks, each containing 80 trials. Half of the participants started with six blocks of salient distractor trials, followed by six blocks of non-salient distractor trials. The order was reversed for the other half of the participants. Each series of six blocks was preceded by 20 practice trials of either salient or non-salient distractor trials, depending on which type of trial they were about to perform. Participants were informed they could take a short break between blocks.

EEG recording and pre-processing

EEG activity was recorded continuously from 64 Ag/AgCl electrodes, positioned according to the 10–20 International system (Sharbrough, 1991), using a Neuroscan Curry 8 system (Compumedics USA, Charlotte, NC, USA) set in AC mode and using an electrode located between FPz and Fz as ground. The vertical electro-oculogram (VEOG) was recorded from two electrodes positioned 1.5 cm above and below the left eye. The horizontal electro-oculogram (HEOG) was recorded from two electrodes positioned on the outer canthi of both eyes. EEG, VEOG, and HEOG signals were band-pass filtered between 0.01 and 40 Hz and digitized at a sampling rate of 1,000 Hz. EEG activity was referenced online to the left earlobe, and then re-referenced offline to the average of the left and right earlobes. Continuous EEG was segmented in epochs starting 100 ms before the visual array onset and ending 600 ms after. EEG epochs were baseline corrected by using the average activity in the time interval -100–0 ms relative to the onset of the search array. After excluding trials associated with an incorrect response in the visual search task, individual trials containing artifacts were then excluded from the final analysis by using the step-function of ERPLAB (step: 30 ms, VEOG deflection > 50 μV within a time window of 150 ms; HEOG deflection > 35 μV within a time window of 200 ms; or signal exceeding ± 100 μV anywhere in the epoch). This procedure led to three subjects being excluded due to more than 30% of trials being rejected. For the remaining 22 subjects, the average percentage of rejected trials was 4.78% (range = 0.8–15%). EEG epochs were then averaged to generate ERPs for each set of salient and non-salient distractor trials.

For laterally displayed targets and distractors, lateralized potential (i.e., Ppc, N2pc, P_D, and SPCN) was computed as the average amplitude from contralateral-minus-ipsilateral difference waves relative to the side of the targets and distractors. Lateralized ERP amplitudes were computed from the averaged activity of P3/4, P5/6, P7/8, PO3/4, P05/6, PO7/8 electrodes, while P3b was measured at electrode Pz. The time windows for estimating the Ppc, N2pc, and SPCN were 100–150 ms, 200–300 ms, and 400–600 ms, respectively, chosen on the basis of previous studies (Berggren & Eimer, 2018; Kerzel & Hyunh Cong, 2022; Jannati et al., 2013). As for the estimation of P_D, a "collapsed localizer" approach was used to avoid biases in selecting the time period for analysis (Luck & Gaspelin, 2017). The time window was defined as the onset and offset of the P_D in the contralateral-minus-ipsilateral different waveform that was averaged across WM-matched and neutral trials (the latencies at which the voltage reached 15% of the peak amplitude). The same approach was also used for the P3b estimation. Thus, the time window used for P_D and P3b was 260–360 ms and 388–530 ms, respectively.

All statistical analyses were performed with R (R Core Team, 2020), using the ezANOVA function of the 'ez' package (Lawrence, 2011). Greenhouse-Geisser correction for non-sphericity was applied when appropriate (Jennings & Wood, 1976), and all comparisons via *t*-test were Bonferroni-corrected.

Results

Behavior

Visual search task

Reaction times (RTs) recorded on trials associated with an incorrect response and/or RTs exceeding three standard deviations above/below individual mean RT (1.4%) were excluded from analysis. A summary of the behavioral results in the visual search task is illustrated in Fig. 2.

Mean RTs and the mean percentage of correct responses were submitted to a 2 × 2 ANOVA considering task type (non-salient distractor vs. salient distractor) and cueing condition (WM-matched vs. neutral) as within-subject factors. RTs were generally shorter on neutral than WM-matched trials (652 vs. 678 ms, F (1, 21) = 106.6, p < .001, η_p^2 = .835). RTs were unaffected by task type, or by the interaction between task type and match type (max F = 1.09, min p = .307). For the analysis of accuracy, participants were more accurate on neutral than WM-matched trials (95.66% vs. 94.52%; F (1, 21) = 13.6, p < .001, η_p^2 = .392). No main effect of task type nor their interaction was found (max F = 1.61, min p = .219).

Memory probe task

Pairwise comparisons revealed that memory probe accuracy did not differ between two tasks (salient distractor task vs. non-salient distractor task: 89.2% vs. 90.4%, t (21) = .9, p = .354).

Event-related potentials

Figure 3 illustrates grand-average contralateral and ipsilateral ERP waveforms. Plotted separately are lateral targets (top panel) and lateral distractors (bottom panel).

Visual inspection of Fig. 3 reveals that lateral targets triggered two salient ERP components, namely the N2pc and the SPCN within the time windows of 200–300 ms and 400–600 ms, respectively. Additionally, it was observed that salient distractors elicited greater contralateral positivity during the Ppc (100–150 ms) and P_D time windows (260–360 ms). The differences in ERPs are more apparent in Fig. 4, which shows the different waveform. These observations were further corroborated by statistical analyses as demonstrated below.

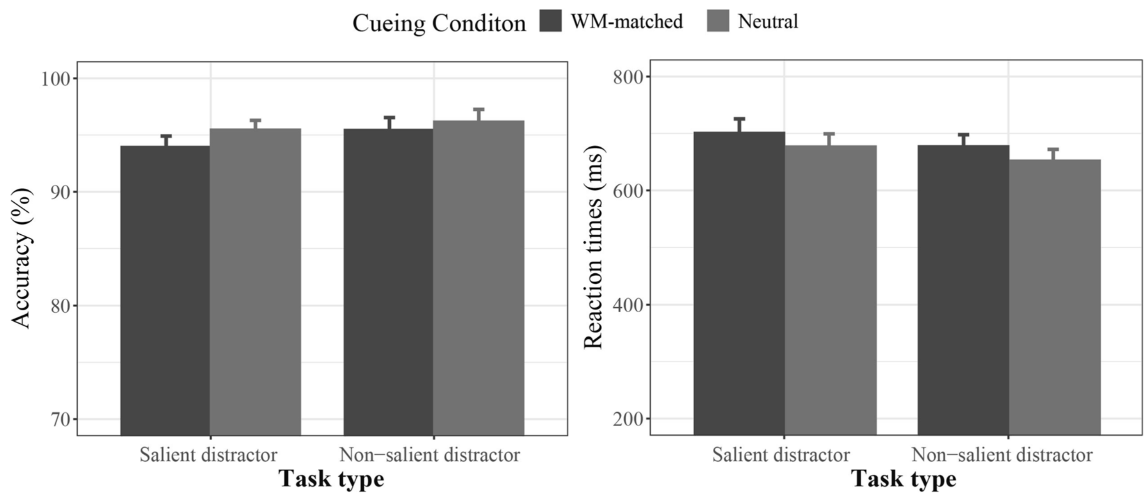

Fig. 2 Mean percentage of correct responses (**left panel**) and mean reaction time (**right panel**) in the visual search task plotted as a function of task type (salient distractor vs. non-salient distractor) and cueing condition (WM-matched vs. neutral). Error bars represent standard error of the mean

N2pc and SPCN for lateral targets

To determine the effect of WM contents on the target's N2pc/SPCN in two tasks, the amplitude values recorded in the N2pc (200–300 ms) and SPCN time-window (400–600 ms) were first separately submitted to a 2 × 2 × 2 repeated-measures ANOVA with task type (non-salient distractor vs. salient distractor), cueing condition (WM-matched vs. neutral), and laterality (contralateral vs. ipsilateral) as within-subject factors.

Analysis of N2pc revealed that there was a main effect of laterality (F (1, 21) = 19.0, p < .001, η_p^2 = .474), indicating the reliable presence of N2pc components (contralateral vs. ipsilateral: M diff = -.36 μV). Furthermore, no other factor effects were found (max F = 2.1, min p = .161), suggesting that N2pc did not differ across task type and cueing condition.

Analysis of SPCN revealed a main effect of cueing condition (F (1, 21) = 12.3, p = .002, η_p^2 = .369), and laterality (F (1, 21) = 20.8, p < .001, η_p^2 = .497). These two effects combined non-linearly (F (1, 21) = 4.9, p = .039, η_p^2 = .188), reflecting the amplitude of SPCN was greater in WM-matched trials than in neutral trials (t (21) = 2.2, p = .039; contralateral vs. ipsilateral: M diff = -.72 μV vs. -.53 μV, respectively).

Ppc, N2pc, and P_D for lateral distractors

With regard to N2pc/SPCN, the amplitude values recorded in the Ppc (100–150 ms) and P_D time-window (260–360 ms) were separately submitted to a 2 × 2 × 2 repeated-measures ANOVA with task type (non-salient distractor vs. salient distractor), cueing condition (WM-matched vs. neutral), and laterality (contralateral vs. ipsilateral) as within-subject factors.

Analysis of Ppc revealed a significant interaction between task type and laterality (F (1, 21) = 11.2, p = .003, η_p^2 = .348). Further planned comparisons showed the presence of a reliable Ppc only in the salient distractor task (contralateral vs. ipsilateral: M diff = .39 μV, p = .005), not in non-salient distractor task (contralateral vs. ipsilateral: M diff = -.12 μV, p = .102). The non-significant three-way interaction (F (1, 21) = .2, p = .637, η_p^2 = .011) further emphasized that the Ppc amplitude did not differ between cueing conditions in the salient distractor task.

On the hypothesis of bottom-up that inherent saliency can affect the efficiency of distractor rejection, the present design allowed us to test whether the P_D amplitude would vary between two tasks. As shown in Fig. 4, cueing condition did not affect P_D amplitude in the salient distractor task, but it did in the non-salient distractor task. This visual impression was confirmed by a significant three-way interaction among task type, cueing condition, and laterality (F (1, 21) = 6.1, p = .022, η_p^2 = .225). Pairwise comparisons showed that in the non-salient distractor task, P_D amplitude was not significant when lateral distractor matched the neutral color (contralateral vs. ipsilateral: M diff = -.21 μV, p = .127), but was clearly present when it matched the WM color (contralateral vs. ipsilateral: M diff = .47 μV, p = .002). In the salient distractor task, the lateral singleton distractor elicited clear P_D amplitude in both WM-matched trials (.56 μV, p = .007) and neutral trials (.52 μV, p < .001). Further planned comparisons of the P_D amplitude in the salient distractor task revealed no amplitude differences between WM-matched trials and neutral trials (.56 μV vs. .52 μV, t (21) = .3, p = .793). Moreover, distractors shared the WM color elicited equivalent P_D in both tasks (non-salient vs. salient: .47 μV vs. .56 μV, t (21) = .4, p = .695).

As it can be seen in Fig. 4, in the salient distractor task, the WM-matched singleton elicited an N2pc at 180–230

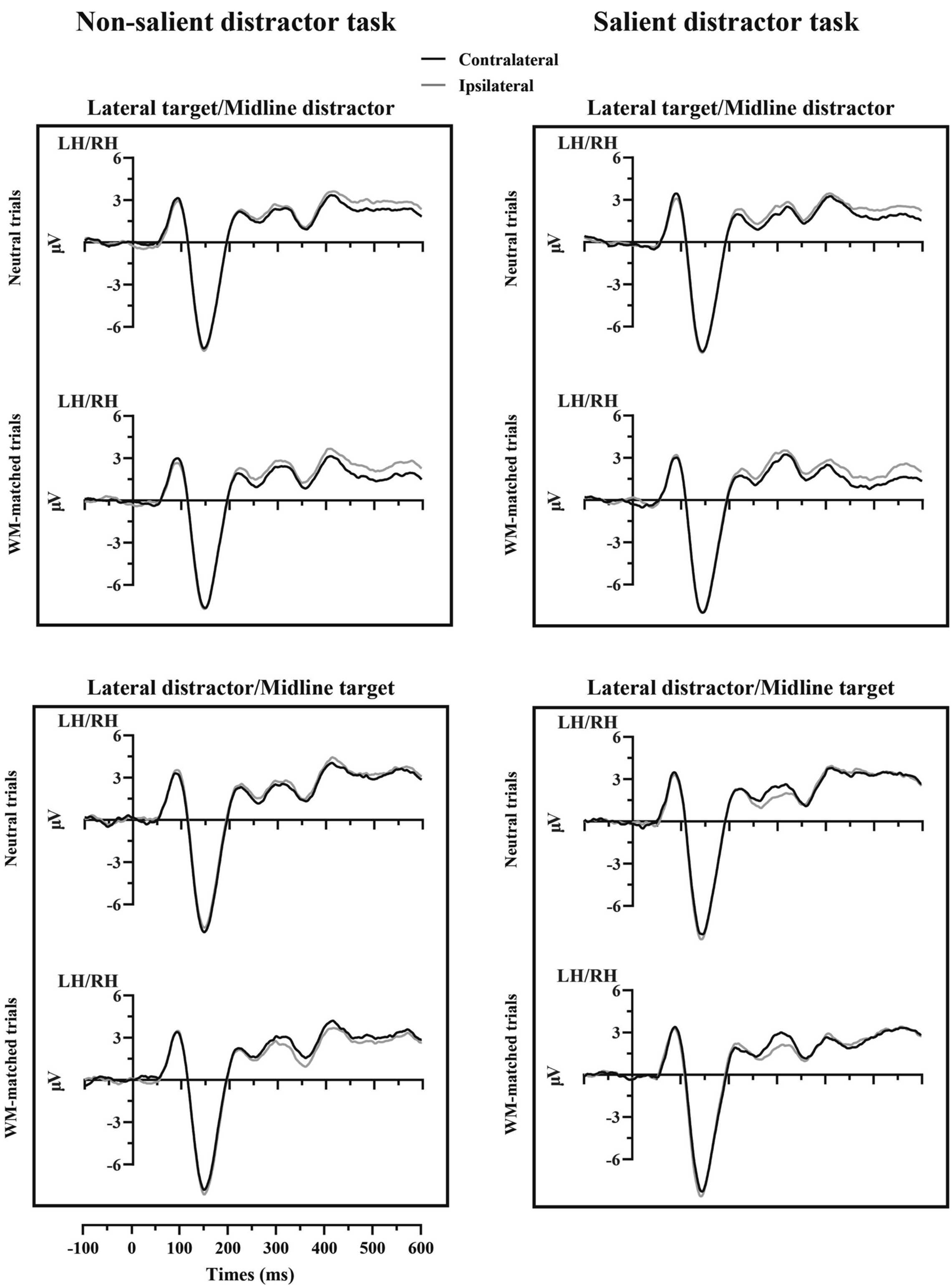

Fig. 3 Event-related potentials (ERPs) plotted separately are lateral targets (**top panel**) and lateral distractors (**bottom panel**), showing the orthogonal combination of the task type (labeled as non-salient distractor on left and salient distractor on right, respectively) and match type (labeled as neutral trials on top and working memory-matched trials on bottom, respectively)

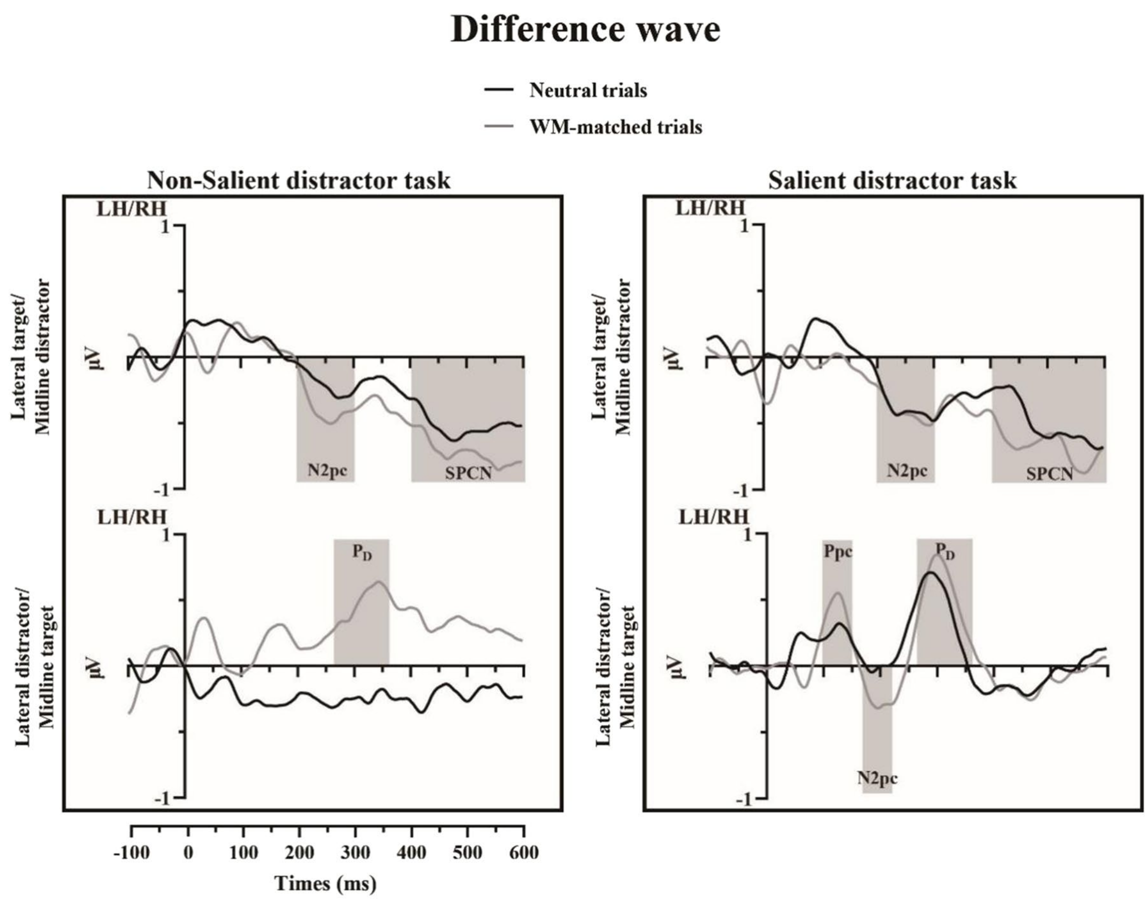

Fig. 4 Different waveforms of event-related potentials (ERPs) for non-salient distractor (**left**) and salient distractor (**right**) trials. Plotted separately are lateral targets (**upper panel**) and lateral distractors (**lower panel**). The areas delimited by the transparent gray rectangles in the graph indicate the time windows considered for ERP amplitude estimation. Lateral targets and lateral distractors recorded on WM-matched trials are represented by gray ERP functions, while black ERP functions represent neutral trials. ERP functions were low-pass filtered at 15 Hz for visualization purposes

ms after visual search array onset. This visual impression was also confirmed by the significant three-way interaction between task type, cueing condition, and laterality (F (1, 21) = 6.9, p = .016, η_p^2 = .246). Pairwise comparisons showed that WM-matched distractor elicited a reliable N2pc in the salient distractor task (contralateral vs. ipsilateral: M diff = -.38 μV, p = .020), but not in the non-salient distractor task (contralateral vs. ipsilateral: M diff = .12 μV, p = .413). In contrast, neutral distractor did not show any N2pc amplitude difference in two tasks (non-salient distractor task: -.19 μV, p = .133; salient distractor task: -.05 μV, p = .700).

P3b for WM-matched and neutral trials

The amplitude values recorded in the P3b time-window (388–530 ms, see also Fig. 5) were then submitted to a repeated-measures ANOVA with task type (non-salient distractor vs. salient distractor) and cueing condition (WM-matched vs. neutral) as within-subject factors. The analysis detected a main effect of cueing condition (F (1, 21) = 15.8, p < .001, η_p^2 = 0.429). Pairwise comparisons confirmed a smaller P3b in WM-matched trials as compared to neutral trials (5.25 μV vs. 5.96 μV). However, the effect of task type appeared to be confined in that P3b was larger in non-salient distractor than in salient distractor tasks (5.98 μV vs. 5.23 μV), but its p-value failed to reach significance (F (1, 21) = 4.1, p = .054, η_p^2 = 0.165).

Discussion

The present study discusses two dominant explanations in the field of visual search regarding the time course of distractor processing, namely the active and reactive suppression accounts. The main point of dispute between these accounts is whether the distractor should attract attention before being rejected. On the behavior level, we found that holding a template for rejection did not optimize visual

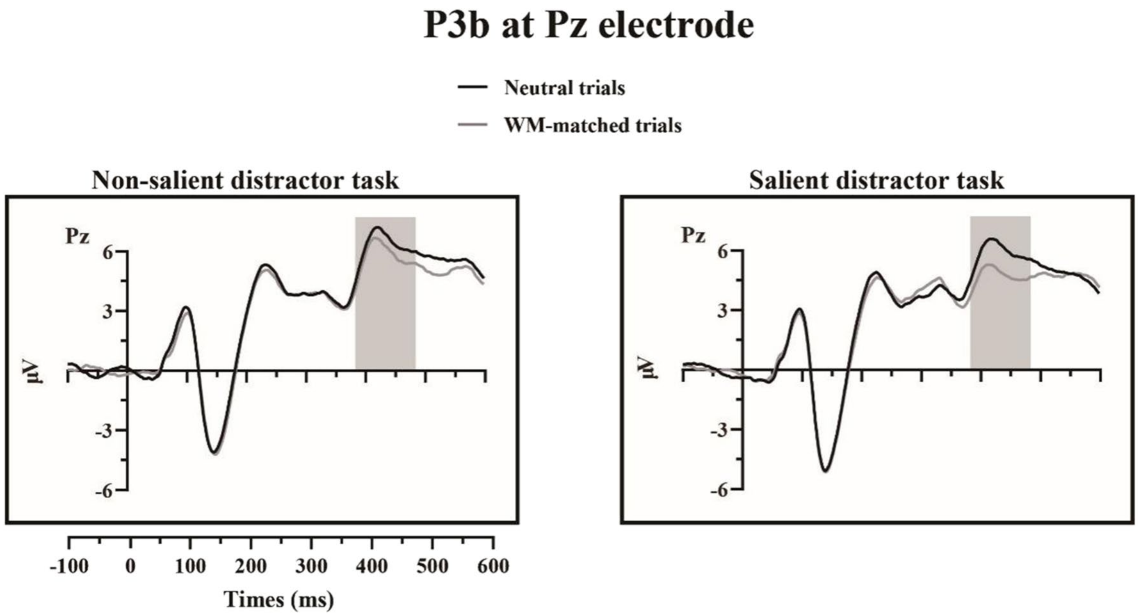

Fig. 5 Event-related potentials (ERPs) elicited at the Pz electrode for non-salient distractors on the left and salient distractors on the right, plotted separately

search, but instead resulted in significantly longer RTs and lower accuracy in tasks where the distractor shared the VWM color. Our study also examined the relationship between saliency and suppression during visual search. Specifically, we investigated whether a singleton distractor (salient distractor task) that matched the color held in VWM was more strongly suppressed than a neutral singleton distractor. Surprisingly, we found that both types of distractors elicited equal amounts of P_D amplitude, suggesting that there was no differential suppression. However, our findings revealed an interesting phenomenon – the WM-matched singleton elicited an N2pc before the onset of the P_D component, indicating that attention initially shifted towards the to-be-ignored location. This result supports our prediction of the reactive suppression account that attention directed towards the WM-matched distractor location prior to its rejection. Our study highlights a discrepancy in suppression between the WM-matched distractor and the bottom-up distractor. Contrary to the notion that greater saliency of a distractor results in more robust suppression in regular visual search, our findings suggest that this principle may not be applicable within the context of WM-driven attention. We provide several opinions on this topic below.

Suppression on the bases of template of rejection

First we consider the finding that equivalent P_D amplitude was found in the salient distractor task for the WM-matched singleton and the neutral singleton. One may argue that this indicates a failed use of the template for rejection when the distractor pop outs, as participants may simply adopt a strategy of suppressing any physically salient object during visual search, irrespective of whether it matches VWM or not. If this is the case, trials in which the WM-matched color reappeared as the singleton distractor should have shown no behavioral modulation if both types of singletons received an equivalent attentional response. However, slower RTs associated with the presence of the WM-matched distractor further confirmed that participants had exhibited an inability to actively suppress the WM-matched distractor.

Despite the target's N2pc remaining unaffected in both search tasks, the SPCN, which represents active maintenance of objects in VWM, was significantly larger for WM-matched trials compared to neutral trials. This suggests a higher memory load in the WM-matched trials, and provides reliable evidence that participants were actively maintaining the distractor color during visual search.

The role of early posterior contralateral positivity

We found that both conditions of salient distractor task elicited a Ppc at 100–150 ms after stimulus onset despite the search display being balanced physically. There has been a surge of interest in the association between specific cognitive processes and the functional significance of this earlier contralateral positivity. Some proposals have suggested that the Ppc primarily indicates perceptual imbalanced saliency across hemifield, independent of attentional processing (Leblanc et al., 2008; Sawaki & Luck, 2010). However, other findings urge researchers to propose that the Ppc reflects attentional processing to some degree (Barras & Kerzel, 2017; Fortier-Gauthier et al., 2012; Kerzel & Hyunh Cong, 2022; Weaver et al., 2017). Chen et al. (2023) showed both an enlarged Ppc and an enlarged P_D for a more salient distractor (stimulus onset) relative to a less salient distractor (color singleton), which could be explained by stronger physical energy for an onset distractor. However, it could also be that the attentional processing of an onset distractor was different from that of a color singleton distractor. Work by Kerzel and Hyunh Cong (2022) found that distractor-elicited P_D, as well as Ppc, varies as a function of the spatial proximity to the target, suggesting the Ppc was sensitive to competition from nearby targets. Furthermore, and in particular related to the present study, where Ppc was found in a visual short-term memory task (Fortier-Gauthier et al., 2012), conclusions from these authors link the Ppc to an initial reactivation of the entire memory representation, and the onset of Ppc manifests itself as a representation of interest. In the context of the signal suppression theory, our finding is also consistent with the idea that the Ppc reflects an "attend-to-me" signal, which may guide subsequent attentional selection or suppression (Chen et al., 2023; Sawaki & Luck, 2010). The behavioral impact of the reappearance of WM items may be explained by earlier and enhanced attentional focusing, reflected in the N2pc or P_D.

We note that the functional significance of the Ppc remains controversial. The study by Sawaki and Luck (2010) directly tested whether this early contralateral positivity is associated with the perception of low-level sensory difference, or is indexed by early suppression. They reduced the attentional resources allocated to peripheral distractors by an embedded central fixation task. With the deeper involvement of attention on the central task, this earlier contralateral positivity was absent. The conclusion from Sawaki and Luck (2010) thereby refutes this early contralateral positivity and merely indexes to perceptual imbalanced saliency, but does reflect some kind of attentional processing (i.e., early suppression). However, a straightforward ascription of the early contralateral positivity to the suppression of distractors is perhaps problematic. While there is an overlap in time windows between the Ppc (100–200 ms) found in our study and the early P_D (115–225 ms) in the study of Sawaki and Luck (2010), we also found a late P_D at 260–360 ms, which is usually and unquestionably interpreted as attentional suppression (Chen et al., 2023; Drisdelle et al., 2023; Hilimire et al. 2011; Liesefeld et al. 2017; Sawaki & Luck, 2013). If we account for the early positivity as attentional suppression, it is hard to explain why there is a later suppression. Does it mean that early suppression is insufficient? This seems problematic to us, and we believe the interpretation of Ppc is more appropriate in the present context.

A recent comprehensive review by Gaspelin et al. (2023) proposed that whether or not the singleton target elicits an early positivity is a key factor in clarifying the functional significance of this earlier contralateral positivity. To obtain this kind of clarity, it is essential to conduct control experiments with a singleton target. Given it was outside the scope of our work, future work may profitably be addressed to investigate this interesting controversy between Ppc and early P_D.

Can the template for rejection boost search efficiency?

When a distractor is present, the increased RT shown in previous literature sometimes has been accounted for as nonspatial filtering costs (Folk & Remington, 1998); for example, the appearance of the to-be-ignored distractor disrupts the processing of target selection without attracting spatial attention to its location (see also Becker, 2007, for a more comprehensive discussion of the relationship between filtering costs and distractor suppression; Wykowska & Schubö, 2011). The aforementioned findings of Kerzel and Hyunh Cong (2022) provide a wonderful example of such a presumption (see also Mazza et al., 2009). They found increased behavioral interference with close distractors, and more importantly, they found the Ppc and P_D components were attenuated when the distractor was close to the target. In line with their prediction based on the biased competition theory – competition between two stimuli increases when they are close together compared to when they are far apart. As such, one would predict that the closer the WM-matched distractor is to the target, the larger the degree of RT cost.

It should be noted that for distractors in the present study, we may not expect the same pattern as Kerzel and Hyunh Cong (2022). Because distractors (gap to the left/right) in the present study did not share the same response with the target (gap to the top/bottom), no response compatibility effect would be expected to occur.[1] Nevertheless, to test this potential account, mean RTs and mean Ppc, as well as P_D amplitudes, were further divided based on the spatial proximity of distractors to the target location. Specifically, on trials when the target was displayed on the midline upper

[1] In the classic flanker task, the target is surrounded on either side by task-irrelevant distractors that can share the same or a different response as the target. The typical result is that RTs are longer when the peripheral distractors are response incongruent than when the flankers are congruent with the response to the centrally presented target. One would account for this slower performance as distractor interference instead of attention captured by peripheral distractors.

hemifield, WM-matched distractors displayed in the lateral lower hemifield are defined as "far," in the lateral upper hemifield is defined as "close," and vice versa for targets displayed on the midline lower hemifield. These data were then submitted to a 2 × 2 × 2 repeated-measures ANOVA, with task type (non-salient distractor vs. salient distractor), cueing condition (WM-matched vs. neutral), and target-distractor distance (close vs. far) as within-subject factors. For the mean RTs, results showed a significant interaction between cueing condition and target-distractor distance (F (1, 21) = 4.65, p = .043, $\eta_p^2 = 0.181$). Interestingly, a further planned comparison revealed that a reliable RT cost was observed to be smaller when the distractor was close to the target than when it was far away (20 vs. 31 ms). The lack of a three-way interaction further emphasized a statistical equivalence of such a spatial proximity modulated RT cost effect was presented in the two tasks ($F = 0.14$, $p = .710$). For the mean Ppc and P_D amplitudes, results yield only the main effect of task type, no other effect was significant (max $F = 2.9$, min $p = .1$). The fact that a faster RT with closer than far WM-match distractors seems inconsistent with those of Kerzel and Hyunh Cong (2022). In an earlier ERP study of Kumar et al. (2009), they found that the N2pc had an earlier onset and greater amplitude when the WM-matched distractor fell on the same side of the display as the target, suggesting facilitated selection of the side of space containing the target when a memory-matching stimulus is also present. We therefore reasoned that it could be that after the detection of the WM-matched distractor, the shift of attention was deployed to the target. The closer the WM-matched distractor is to the target, the faster this shift of attention occurs. This presumption is similar to the "search-and-destroy" account described by Moher and Egeth (2012); though no sign of evidence has shown that attention must be captured by the WM-matched distractor (which is discussed below), analysis of behavioral data reveals that spatial proximity between the target and distractor does mediate this process. The above result lends further support for the notion that distractor rejection cannot be achieved by explicitly activating the foreknowledge of distractor. As proposed by Beck and Hollingworth (2015), it may be difficult to implement a rejection template without an efficient way to spatially recode the search array.

Is memory-driven attentional capture contingent on display heterogeneity?

The absence of an N2pc for the WM-matched distractor in the non-salient distractor task suggests it failed to capture attention, which raises a further possibility: memory-driven capture is contingent on goal-driven control in this case. If the degree to which the distractor captures attention is affected by display heterogeneity, we would at least expect a greater RT cost for the WM-matched distractors in the salient distractor task than in the non-salient distractor task, since the former is more "pop-out". Nevertheless, behavioral data yield no significant interaction between search task and cueing condition, indicating the statistical equivalence of RT cost between two tasks. In this regard, our findings were consistent with those of Olivers (2009, Experiment 2), as the RTs cost in that study did not differ between homogeneous and heterogeneous displays. Therefore, we cannot simply conclude that WM-matched distractors in the non-salient distractor tasks fail to capture attention due to the absence of N2pc.

To gain insight into the stages of information processing during the search task with WM-matched distractors, we then collapsed the data based on the match type (WM-match vs. neutral). Results showed that the parietal P3b was attenuated for WM-matched relative to neutral trials. P3b in the classical rapid serial visual presentation paradigm is also a typical ERP signature of the attentional blink, varied with intertarget lag, with diminished amplitude and postponed latency at short relative to long lags (Dell'acqua et al., 2015; Dell'acqua et al., 2016). In the present context of visual search, P3b can at least reflect the consolidation of visual information in VWM after the completion of perception (Akyürek et al., 2010; Cosman et al., 2016). Previous studies agree on the prospect that selective attention and WM share limited resources, resulting in a competitive interaction between the two (Lavie, 2005; Mayer et al., 2007; Soto et al., 2007; for a review, see Kiyonaga & Egner, 2013). Since the N2pc elicited by lateral targets was similar for both match types, we suggest that foreknowledge of the distractor did not affect early target detection but hindered later target discrimination. Under the assumption of holding a "template for rejection" to filter distractor, the WM-matched distractor in most cases wins the competition for external attentional processing. Although such processing may bolster the representation of that item (which we cannot ascertaom in the present study because the memory probe task was not given following the visual search task), the processing of other task-relevant stimuli will suffer. That is, the competition between the target processing and the WM-matched distractor rejection impacts the search performance.

The "search-and-destroy" account under the present study

Further to our interpretation of the "search-and-destroy" account – that is, whether such rejection requires us to deploy our attention to the to-be-ignored object's location, in an eye-tracking study, Beck et al. (2018) found that rejection was not contingent on initial capture by the WM-matched distractor. In the present study, the fact that only the WM-matched singleton, not the neutral singleton, elicited the N2pc suggested that suppression towards the

WM-matched distractor is different from that of physical salient distractors. Obviously, the operation of a template for rejection depended on directing attention to the to-be-ignored item. However, due to the lack of a truly "baseline" condition in the salient distractor task (i.e., trials without any singleton distractor), we were unable to conclude whether the neutral singleton was actively suppressed. After all, the main purpose of the present study was to examine whether reducing display heterogeneity would lead to more robust suppression. Future work will need to clarify this possibility by including a condition where the singleton distractor is absent.

Recently, Carlisle and Nitka (2019) have suggested the above location-based strategy may be involved in distractor rejection. Using very similar procedures, but the color of search items was segregated by hemifield (e.g., blue items on the left, red on the right), half of the items in the display matched either a positive cue or a negative cue that was given ahead of the search task. They proposed that the "search and destroy" account would predict a serial shift of attention in which participants first attend the hemifield containing the cued distractors, followed by attention being directed to the hemifield containing the potential targets, as the ERP waveform would show a positivity followed by negativity contralateral to the target side in the negative cue condition. They found the lateral target only elicited N2pc, and therefore concluded that the template for rejection operates in an active suppression manner. However, we wonder if their design was able to examine such a serial shift of attention. One factor that might account for this discrepancy is their search target and distractor were always presented bilaterally in different fields, which created inherent ambiguity because a change in N2pc, say an increase in amplitude, could be due to a negative attentional response recorded on the contralateral portion, positive attentional response on ipsilateral portion, or both. When only the target is displayed in the lateral position, it is clear that the N2pc is the recorded activation difference from which a predominantly contralateral portion contributes. But when the WM-matched distractor is displayed opposite to the target, the classic contra-minus-ipsi approach may likely subtract the attentional response to the distractor, which is now contralateral to it but on the ipsilateral side of the target. Recent work from Monnier et al. (2020) showed that attentional responses to lateral displayed objects were observed in both contralateral and ipsilateral portions, with the ipsilateral response onset slightly later than the contralateral response. Consequently, even though Carlisle and Nitka (2019) eventually found only the N2pc to the target side, they were unable to determine whether the contralateral negativity would be subtracted from an unknown ipsilateral response, as their N2pc in the negative cue condition was delayed and smaller compared to that of N2pc in the positive cue condition. In the present study, the target and distractor were controlled by a pseudorandom order, with one placed along the vertical midline while the other in the lateral position. Midline stimuli have little contribution to the lateralized activity due to the lateralization calculation (Woodman & Luck, 2003), thus enabling a more precise interpretation of observed lateralized activity to the processing of the lateral item. For this reason, decomposing a lateralized attentional response to target and WM-matched distractor is deemed practical in our case.

It is also informative to compare our results with those of Berggren and Eimer (2021). They found that RT cost dissipated only after practice with the same negative cue for approximately 25–50 trials (Experiment 3) and eventually turned into benefits after several hundreds of trials (Experiment 4). This is in line with our findings in the non-salient distractor task, in which the distractor shared the color of the negative cue had an impact on the search performance, while this lateral WM-matched distractor only elicited the suppression-related P_D component. Conclusions from Berggren and Eimer (2021) suggest that holding the distractor feature ahead of visual search cannot optimize the attentional selection, but interference. And the benefit from the constant negative cue cannot account for the active suppression but is "*likely to be the result of passive habituation processes*". However, it is important to note that their negative cue remained constant across blocks, which may lead to the control of the template for rejection being governed by long-term memory (LTM, Carlisle et al., 2011; van Moorselaar et al., 2016) rather than by WM, as in the present study (see also Kugler et al., 2015). Previous research has shown that repeatedly being exposed to the same stimuli leads to strong attenuation of behavioral and neural responses (Bell et al., 2012; Gunseli et al., 2014). For example, Turatto and Pascucci (2016) have shown that attentional capture by the salient distractor is subject to habituation. With enough practice, there was no effect of a high-luminance peripheral onset in the visual discrimination task, manifesting as attentional selection becomes fully immune to LTM-matched distractors. Thus, more work is needed to elucidate attentional suppression from the LTM template.

Overall, the results of the current study suggest that the implementation of the template for rejection is reactive. When attention is deployed to the WM-matched distractor, rejection occurs after it has been identified as a non-target. Under heterogeneous conditions (non-salient distractor task), it is not always necessary for distractor rejection to be accompanied by attentional capture. However, rejecting the WM-matched distractor can interfere with target discrimination. Given the fact that we fail to find any RT benefit from foreknowledge of a distractor, it may turn out that implementing the template for rejection proactively is extremely exacting (for a more detailed review, see Luck et al., 2021). In other studies, time intervals seem to be the factor to form a solid template for rejection

(Han & Kim, 2009). For example, In the study of Tanda and Kawahara (2019), they found an RT benefit when the stimulus onset asynchrony (SOA) between the cue display and search display was approximately 2,400 ms or longer. It appears to be that our time interval (SOA: 1,100 ms) was too short to configure a fully fledged template for rejection. However, a similar conclusion was drawn by Beck et al. (2018). They consistently found distractors that matched the to-be-ignored color capture first fixation more, but this ocular capture effect was then reduced when given a longer SOA (more than 1,000 ms). Their conclusions suggest that processes toward the to-be-ignored distractor include the initial attentional capture (which results in RT cost) and the later avoidance (which causes a RT benefit). Whether the end-of-trial manual responses record a benefit or cost depends on the relative differences in the magnitudes of early capture costs and later avoidance benefits (see also Becker et al., 2015).

Conclusions

To summarize, this study showed that the degree of WM-matched distractor suppression was not affected by display heterogeneity, but the attentional capture was. Our results have provided evidence that holding the distractor feature actively in WM first results in attentional capture, followed by attentional suppression. Additionally, the target surrounded by either WM-matched or neutral distractors elicited identical N2pc, but the diminished P3b suggested that target discrimination efficiency was disrupted. Taken together, we propose that the template for rejection appears to operate reactively.

Acknowledgements This study was supported by grants from the National Natural Science Foundation of China (No. 31970993 and No. 32271107) to SF. Data and materials for the experiments are available upon request from the authors. The experiment was not preregistered.

Data Availability We have no plans to submit data availability statements. Data and materials for the experiments are available upon request from the authors.

References

Addleman, D. A., & Störmer, V. S. (2022). No evidence for proactive suppression of explicitly cued distractor features. *Psychonomic Bulletin & Review*, *29*(4), 1338–1346.

Akyürek, E. G., Leszczyński, M., & Schubö, A. (2010). The temporal locus of the interaction between working memory consolidation and the attentional blink. *Psychophysiology, 47*(6), 1134–1141.

Arita, J. T., Carlisle, N. B., & Woodman, G. F. (2012). Templates for rejection: Configuring attention to ignore task-irrelevant features. *Journal of Experimental Psychology: Human Perception and Performance, 38*(3), 580–584.

Bacon, W. F., & Egeth, H. E. (1994). Overriding stimulus-driven attentional capture. *Perception & Psychophysics, 55*(5), 485–496.

Barras, C., & Kerzel, D. (2017). Salient-but-irrelevant stimuli cause attentional capture in difficult, but attentional suppression in easy visual search. *Psychophysiology, 54*, 1826–1838.

Beck, V. M., & Hollingworth, A. (2015). Evidence for negative feature guidance in visual search is explained by spatial recoding. *Journal of Experimental Psychology: Human Perception and Performance, 41*(5), 1190–1196.

Beck, V. M., Luck, S. J., & Hollingworth, A. (2018). Whatever you do, don't look at the...: Evaluating guidance by an exclusionary attentional template. *Journal of Experimental Psychology: Human Perception and Performance, 44*(4), 645–662.

Becker, S. I. (2007). Irrelevant singletons in pop-out search: Attentional capture or filtering costs? *Journal of Experimental Psychology: Human Perception & Performance, 33*(4), 764–787.

Becker, M. W., Hemsteger, S., & Peltier, C. (2015). No templates for rejection: A failure to configure attention to ignore task-irrelevant features. *Visual Cognition, 23*(9–10), 1150–1167.

Bell, R., Röer, J. P., Dentale, S., & Buchner, A. (2012). Habituation of the irrelevant sound effect: Evidence for an attentional theory of short-term memory disruption. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 38*(6), 1542–1557.

Berggren, N., & Eimer, M. (2018). Object-based target templates guide attention during visual search. *Journal of Experimental Psychology: Human Perception and Performance, 44*(9), 1368–1382.

Berggren, N., & Eimer, M. (2021). The guidance of attention by templates for rejection during visual search. *Attention, Perception, & Psychophysics, 83*(1), 38–57.

Burra, N., & Kerzel, D. (2013). Attentional capture during visual search is attenuated by target predictability: Evidence from the N2pc, Pd, and topographic segmentation. *Psychophysiology, 50*(5), 422–430.

Carlisle, N. B. (2023). Negative and positive templates: Two forms of cued attentional control. *Attention, Perception, & Psychophysics, 85*(3), 585–595.

Carlisle, N. B., & Nitka, A. W. (2019). Location-based explanations do not account for active attentional suppression. *Visual Cognition, 27*(3–4), 305–316.

Carlisle, N. B., Arita, J. T., Pardo, D., & Woodman, G. F. (2011). Attentional templates in visual working memory. *Journal of Neuroscience, 31*(25), 9315–9322.

Chen, X., Xu, B., Chen, Y., Zeng, X., Zhang, Y., & Fu, S. (2023). Saliency affects attentional capture and suppression of abrupt-onset and color singleton distractors: Evidence from event-related potential studies. *Psychophysiology, 60*, e14290.

Conci, M., Deichsel, C., Müller, H. J., & Töllner, T. (2019). Feature guidance by negative attentional templates depends on search difficulty. *Visual Cognition, 27*(3–4), 317–326.

Cosman, J. D., Arita, J. T., Ianni, J. D., & Woodman, G. F. (2016). Electrophysiological measurement of information flow during visual search. *Psychophysiology, 53*(4), 535–543.

Dell'Acqua, R., Dux, P. E., Wyble, B., Doro, M., Sessa, P., Meconi, F., & Jolicœur, P. (2015). The attentional blink impairs detection and delays encoding of visual information: Evidence from human electrophysiology. *Journal of Cognitive Neuroscience, 27*, 720–735.

Dell'Acqua, R., Doro, M., Dux, P. E., Losier, T., & Jolicour, P. (2016). Enhanced frontal activation underlies sparing from the attentional blink: Evidence from human electrophysiology. *Psychophysiology, 53*(5), 623–633.

Desimone, R., & Duncan, J. (1995). Neural mechanisms of selective visual attention. *Annual Review of Neuroscience, 18*(1), 193–222.

Dowd, E. W., Kiyonaga, A., Egner, T., & Mitroff, S. R. (2015). Attentional guidance by working memory differs by paradigm: An individual-differences approach. *Attention, Perception, & Psychophysics, 77*(3), 704–712.

Drisdelle, B. L., Corriveau, I., Fortier-Gauthier, U., & Jolicoeur, P. (2023). Task-irrelevant filler items alter the dynamics of electrical brain activity during visual search. *Quarterly Journal of Experimental Psychology, 76*(6), 1245–1263.

Duncan, J., & Humphreys, G. (1992). Beyond the Search Surface. *Journal of Experimental Psychology Human Perception & Performance, 18*(2), 578–588.

Eimer, M. (1996). The N2pc component as an indicator of attentional selectivity. *Electroencephalography and Clinical Neurophysiology, 99*(3), 225–234.

Folk, C. L., & Remington, R. (1998). Selectivity in distraction by irrelevant featural singletons: Evidence for two forms of attentional capture. *Journal of Experimental Psychology: Human Perception and Performance, 24*(3), 847–858.

Folk, C. L., Remington, R. W., & Johnston, J. C. (1992). Involuntary covert orienting is contingent on attentional control settings. *Journal of Experimental Psychology: Human Perception and Performance, 18*(4), 1030–1044.

Fortier-Gauthier, U., Moffat, N., Dell'Acqua, R., McDonald, J. J., & Jolicœur, P. (2012). Contralateral cortical organisation of information in visual short-term memory: Evidence from lateralized brain activity during retrieval. *Neuropsychologia, 50*(8), 1748–1758.

Gaspelin, N., & Luck, S. J. (2018a). Combined electrophysiological and behavioral evidence for the suppression of salient distractors. *Journal of Cognitive Neuroscience, 30*(9), 1265–1280.

Gaspelin, N., & Luck, S. J. (2018b). The role of inhibition in avoiding distraction by salient stimuli. *Trends in Cognitive Sciences, 22*(1), 79–92.

Gaspelin, N., & Luck, S. J. (2019). Inhibition as a potential resolution to the attentional capture debate. *Current Opinion in Psychology, 29*, 12–18.

Gaspelin, N., Lamy, D., Egeth, H. E., Liesefeld, H. R., Kerzel, D., Mandal, A., … van Moorselaar, D. (2023). The distractor positivity component and the inhibition of distracting stimuli. *Journal of Cognitive Neuroscience*, *35*(11), 1693–1715.

Gunseli, E., Olivers, C. N., & Meeter, M. (2014). The handoff of the attentional template from working memory after repeated search: The effects of task difficulty. *Journal of Vision, 14*(10), 712–712.

Han, S. W., & Kim, M.-S. (2009). Do the contents of working memory capture attention? Yes, but cognitive control matters. *Journal of Experimental Psychology: Human Perception and Performance, 35*(5), 1292–1302.

Hickey, C., Di Lollo, V., & McDonald, J. J. (2008). Target and distractor processing in visual search: Decomposition of the N2pc. *Visual Cognition, 16*(1), 110–113.

Hilimire, M. R., Mounts, J. R. W., Parks, N. A., & Corballis, P. M. (2011). Dynamics of target and distractor processing in visual search: Evidence from event-related brain potentials. *Neuroscience Letters, 495*(3), 196–200.

Jannati, A., Gaspar, J. M., & McDonald, J. J. (2013). Tracking target and distractor processing in fixed-feature visual search: Evidence from human electrophysiology. *Journal of Experimental Psychology: Human Perception and Performance, 39*, 1713–1730.

Jennings, J. R., & Wood, C. C. (1976). The e-adjustment procedure for repeated-measures analyses of variance. *Psychophysiology, 13*(3), 277–278.

Kerzel, D., & Hyunh Cong, S. (2022). Biased competition between targets and distractors reduces attentional suppression: Evidence from the positivity posterior contralateral and distractor positivity. *Journal of Cognitive Neuroscience, 34*(9), 1563–1575.

Kiyonaga, A., & Egner, T. (2013). Working memory as internal attention: Toward an integrative account of internal and external selection processes. *Psychonomic Bulletin & Review, 20*, 228–242.

Kugler, G., 't hart, B. M., Kohlbecher, S., Einhäuser, W., & Schneider, E. (2015). Gaze in visual search is guided more efficiently by positive cues than by negative cues. *PloS One, 10*(12), e0145910.

Kumar, S., Soto, D., & Humphreys, G. W. (2009). Electrophysiological evidence for attentional guidance by the contents of working memory. *European Journal of Neuroscience, 30*(2), 307–317.

Lavie, N. (2005). Distracted and confused?: Selective attention under load. *Trends in Cognitive Sciences, 9*(2), 75–82.

Lawrence, M. A. (2011). ez: Easy analysis and visualization of factorial experiments. *Computer Software Manual (R Package Version 3.0- 0).*

Leber, A. B., & Egeth, H. E. (2006). It's under control: Top-down search strategies can override attentional capture. *Psychonomic Bulletin & Review, 13*(1), 132–138.

Leblanc, É., Prime, D. J., & Jolicoeur, P. (2008). Tracking the location of visuospatial attention in a contingent capture paradigm. *Journal of Cognitive Neuroscience, 20*(4), 657–671.

Liesefeld, H. R., Liesefeld, A. M., Töllner, T., & Müller, H. J. (2017). Attentional capture in visual search: Capture and post-capture dynamics revealed by EEG. *NeuroImage, 156*, 166–173.

Luck, S. J., & Gaspelin, N. (2017). How to get statistically significant effects in any ERP experiment (and why you shouldn't). *Psychophysiology, 54*(1), 146–157.

Luck, S. J., & Hillyard, S. A. (1994). Spatial filtering during visual search: Evidence from human electrophysiology. *Journal of Experimental Psychology: Human Perception and Performance, 20*(5), 1000–1014.

Luck, S. J., Gaspelin, N., Folk, C. L., Remington, R. W., & Theeuwes, J. (2021). Progress toward resolving the attentional capture debate. *Visual Cognition, 29*(1), 1–21.

Mayer, J. S., Bittner, R. A., Nikolić, D., Bledowski, C., Goebel, R., & Linden, D. E. (2007). Common neural substrates for visual working memory and attention. *NeuroImage, 36*(2), 441–453.

Mazza, V., Turatto, M., & Caramazza, A. (2009). Attention selection, distractor suppression and N2pc. *Cortex, 45*(7), 879–890.

Moher, J., & Egeth, H. E. (2012). The ignoring paradox: Cueing distractor features leads first to selection, then to inhibition of to-be-ignoredd items. *Attention, Perception, & Psychophysics, 74*(8), 1590–1605.

Monnier, A., Dell'Acqua, R., & Jolicoeur, P. (2020). Distilling the salient contralateral and ipsilateral attentional responses to lateral stimuli and the bilateral response to midline stimuli for upper and lower visual hemifield locations. *Psychophysiology, 57*(11), e13651.

Olivers, C. N. (2009). What drives memory-driven attentional capture? The effects of memory type, display type, and search type. *Journal of Experimental Psychology: Human Perception and Performance, 35*(5), 1275–1291.

Peters, J. C., Roelfsema, P. R., & Goebel, R. (2008). Selective attentional guidance by items in working memory: converging fMRI and ERP results. *NeuroImage, 41*(s96). https://cris.maastrichtuniversity.nl/en/publications/selective-attentional-guidance-by-items-in-working-memory-converg

Polich, J. (2007). Updating P300: An integrative theory of P3a and P3b. *Clinical Neurophysiology, 118*(10), 2128–2148.

R Core Team (2020) R: A Language and Environment for Statistical Computing. *R Foundation for Statistical Computing*, Vienna, Austria. https://scholar.google.com/scholar?q=R%20Core%20Team%2C%202020.%20R%3A%20A%20language%20and%20environment%20for%20statistical%20computing

Rac-Lubashevsky, R., & Kessler, Y. (2019). Revisiting the relationship between the P3b and working memory updating. *Biological Psychology, 148*, 107769.

Sawaki, R., & Luck, S. J. (2010). Capture versus suppression of attention by salient singletons: Electrophysiological evidence for an automatic attend-to-me signal. *Attention, Perception, & Psychophysics, 72*(6), 1455–1470.

Sawaki, R., & Luck, S. J. (2013). Active suppression after involuntary capture of attention. *Psychonomic Bulletin & Review, 20*(2), 296–301.

Sharbrough, F. (1991). American Electroencephalographic Society guidelines for standard electrode position nomenclature. *Journal of Clinical Neurophysiology, 8*(2), 200–202.

Soto, D., Humphreys, G. W., & Rotshtein, P. (2007). Dissociating the neural mechanisms of memory-based guidance of visual selection. *Proceedings of the National Academy of Sciences of the United States of America, 104*(43), 17186–17191.

Tanda, T., & Kawahara, J. (2019). Association between cue lead time and template-for-rejection effect. *Attention, Perception & Psychophysics, 81*(6), 1880–1889.

Theeuwes, J. (2010). Top–down and bottom–up control of visual selection. *Acta Psychologica, 135*(2), 77–99.

Turatto, M., & Pascucci, D. (2016). Short-term and long-term plasticity in the visual-attention system: Evidence from habituation of attentional capture. *Neurobiology of Learning and Memory, 130*, 159–169.

van Moorselaar, D., Theeuwes, J., & Olivers, C. N. (2016). Learning changes the attentional status of prospective memories. *Psychonomic Bulletin & Review, 23*(5), 1483–1490.

Vogel, E. K., & Luck, S. J. (2002). Delayed working memory consolidation during the attentional blink. *Psychonomic Bulletin & Review, 9*(4), 739–743.

Weaver, M. D., van Zoest, W., & Hickey, C. (2017). A temporal dependency account of attentional inhibition in oculomotor control. *NeuroImage, 147*, 880–894.

Wolfe, J. M. (2012). Saved by a log: How do humans perform hybrid visual and memory search? *Psychological Science, 23*(7), 698–703.

Woodman, G. F., & Luck, S. J. (2003). Serial deployment of attention during visual search. *Journal of Experimental Psychology: Human Perception and Performance, 29*(1), 121–138.

Woodman, G. F., & Luck, S. J. (2007). Do the contents of visual working memory automatically influence attentional selection during visual search? *Journal of Experimental Psychology: Human Perception and Performance, 33*(2), 363–377.

Wykowska, A., & Schubö, A. (2011). Irrelevant singletons in visual search do not capture attention but can produce nonspatial filtering costs. *Journal of Cognitive Neuroscience, 23*(3), 645–660.

Zhang, Z., & Carlisle, N. B. (2023). Assessing recoding accounts of negative attentional templates using behavior and eye tracking. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 49*(4), 509.

Zhang, W., & Luck, S. J. (2008). Discrete fixed-resolution representations in visual working memory. *Nature, 453*(7192), 233–235.

Zhang, Z., Gaspelin, N., & Carlisle, N. B. (2020). Probing early attention following negative and positive templates. *Attention, Perception, & Psychophysics, 82*(3), 1166–1175.